Emergence of Statistical-Ensemble Equivalence in a Two-Dimensional Interacting Quantum Lattice Gas

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A computational statistical-mechanics study by exact diagonalization

Abstract. We study the equivalence of the microcanonical, canonical, and grand-canonical ensembles in a two-dimensional hard-core Bose–Hubbard model—an interacting *quantum lattice gas*—as a controlled function of system size and interaction strength. Using full-spectrum exact diagonalization on square lattices of $L \times L = 4, 9, 16$ sites, in every conserved particle-number sector, we compute each ensemble's thermodynamics from a single shared set of eigenvalues. We find that finite-difference microcanonical temperature and chemical potential are numerically ill-conditioned on the small bounded discrete spectra reachable by exact diagonalization, and we therefore quantify ensemble equivalence through well-conditioned fluctuation observables: the relative energy fluctuation $\sigma_E/|\langle E\rangle|$ (microcanonical↔canonical) and the relative number fluctuation $\sigma_N/\langle N\rangle$ (canonical↔grand-canonical), together with the per-site canonical–grand energy gap. For the non-interacting ($V = 0$) and repulsive ($V/t = +2$) gases, all three discrepancy metrics decrease monotonically with system size, following approximate power laws in $M = L^2$ and demonstrating the expected emergence of equivalence. For attractive interactions the relative number fluctuation fails to shrink, and for strong attraction ($V/t = -3$) its finite-size exponent turns *positive* ($\sim M^{+0.13}$, reaching 0.96 at $M = 16$): the canonical–grand mismatch *grows* with system size, signalling the clustering/phase-separation instability of an attractive lattice gas. Sweeping the interaction over V/t from -3 to $+3$ reveals this as a smooth, monotonic trend in the number channel.

1 Introduction

The equivalence of statistical ensembles is a foundational assertion of equilibrium statistical mechanics: in the thermodynamic limit, the microcanonical (N, V, E), canonical (N, V, T), and grand-canonical (μ, V, T) descriptions yield the same intensive thermodynamics. It is a *limit* statement. For finite systems—nanoscale devices, cold-atom quantum simulators, and all numerically exact many-body calculations—the ensembles differ by corrections that vanish only as the particle number grows. Understanding the rate and mechanism of this convergence, and when it fails, is both conceptually central and practically important.

We address the question in a minimal but nontrivial quantum model and ask: how does the equivalence of the three ensembles emerge in an interacting quantum lattice gas as system size increases, and how do quantum hopping and interaction affect the finite-size discrepancies? We study the two-dimensional hard-core Bose–Hubbard model—a quantum lattice gas in the statistical-mechanical sense (indistinguishable quantum particles on lattice sites with tunnelling and interaction), realized by ultracold bosons in optical lattices in the hard-core limit. This is unrelated to the identically named "quantum lattice-gas automaton" of quantum computing; we study the many-body physical model.

Our methodology is correctness-first: we build the Hamiltonian in each conserved N-sector, validate against independent analytic and brute-force results, diagonalize fully, and only then compare ensembles. A central methodological finding—that finite-difference microcanonical T and μ are unreliable at these sizes—leads us to anchor the study on fluctuation observables that are smooth sums over the spectrum.

2 Model and methods

2.1 Hamiltonian

On an $L \times L$ square lattice with $M = L^2$ sites we consider hard-core bosons with nearest-neighbor hopping and interaction,

$$\hat{H} = -t \sum_{\langle i,j \rangle} \left(\hat{b}_i^\dagger \hat{b}_j + \hat{b}_j^\dagger \hat{b}_i \right) + V \sum_{\langle i,j \rangle} \hat{n}_i \hat{n}_j, \quad \hat{n}_i = \hat{b}_i^\dagger \hat{b}_i, \quad (1)$$

with the hard-core constraint $\hat{n}_i \in \{0, 1\}$, $\hat{b}_i^2 = (\hat{b}_i^\dagger)^2 = 0$. Here t is the hopping amplitude, V the nearest-neighbor interaction ($V > 0$ repulsive, $V < 0$ attractive). We set $k_B = 1$ and measure energy in units of t, so the controls are V/t, T/t, μ/t, and use periodic boundary conditions. The Hamiltonian conserves particle number, $[\hat{H}, \hat{N}] = 0$, hence is block-diagonal in N: we diagonalize each fixed-N block and reuse the eigenvalues in all three ensembles. In the occupation basis hard-core bosons carry no inter-site exchange sign, so each hopping element is $-t$ and every block is real symmetric.

2.2 Basis and exact diagonalization

A basis state is an occupation string encoded as an integer whose i-th bit is n_i. The full space has dimension 2^M; the fixed-N sector has dimension $\binom{M}{N}$. Because the

canonical and grand partition functions sum over the *entire* spectrum, we require all eigenvalues of each block and use a dense symmetric eigensolver (`eigvalsh`) rather than an iterative method. The ceiling is the half-filling sector $\binom{M}{M/2}$; for $L = 4$ this is 12 870, diagonalized in minutes. Per-sector spectra are cached once and reused.

2.3 Three ensembles from one spectrum

With $E_{N,\alpha}$ the α-th eigenvalue of sector N: microcanonical $\Omega(E, N; \Delta E) = \#\{\alpha : E \le E_{N,\alpha} < E + \Delta E\}$, $S = \ln \Omega$; canonical $Z_N = \sum_\alpha e^{-\beta E_{N,\alpha}}$, $\langle E \rangle_N = -\partial_\beta \ln Z_N$, $C_V = \beta^2 \mathrm{Var}(E)$; grand-canonical $\Xi = \sum_N e^{\beta \mu N} Z_N$, $\langle N \rangle = \sum_{N,\alpha} N p_{N,\alpha}$, $\partial_\mu \langle N \rangle = \beta \mathrm{Var}(N)$, with $p_{N,\alpha} = e^{-\beta(E_{N,\alpha} - \mu N)}/\Xi$. All partition functions use a log-sum-exp shift to prevent overflow.

2.4 Measuring equivalence: fluctuation metrics

Extracting T, μ from microcanonical finite differences is fragile on small bounded discrete spectra: $S(E) = \ln \Omega$ is a jagged window-dependent histogram whose derivative diverges in spectral gaps (low T) and flips sign near the density-of-states maximum (the negative-temperature region of a bounded spectrum). We verified this across surface-entropy, Gaussian-broadened, and Gibbs/volume-entropy variants; it is an intrinsic small-system limitation.

We therefore quantify equivalence through smooth spectral sums requiring no differentiation:

$$r_E \equiv \frac{\sigma_E}{|\langle E \rangle|} = \frac{\sqrt{\mathrm{Var}(E)}}{|\langle E \rangle|} \sim M^{-1/2}, \tag{2}$$

$$r_N \equiv \frac{\sigma_N}{\langle N \rangle} = \frac{\sqrt{\mathrm{Var}(N)}}{\langle N \rangle} \sim M^{-1/2}, \tag{3}$$

$$\Delta_E / M \equiv \frac{1}{M} \left| \langle E \rangle_{\mathrm{grand}} - \langle E \rangle_{\mathrm{can}} \right|. \tag{4}$$

r_E measures the sharpness of the canonical energy saddle (micro↔canonical); r_N measures the number-distribution width at μ^* tuned so $\langle N \rangle = N$ (canonical↔grand). Since $\langle N \rangle(\mu)$ is monotone ($\partial_\mu \langle N \rangle = \beta \mathrm{Var}(N) > 0$), μ^* is unique and found by Brent's method to 10^{-10}. Finite-difference T, μ are kept only as a noisy secondary diagnostic.

3 Validation

The implementation was validated against independent references, all encoded as automated tests: (i) Hermiticity of every block to machine precision; (ii) sector dimension $\binom{M}{N}$ with sectors partitioning 2^M; (iii) particle-number conservation—an independently built full-space Hamiltonian is block-diagonal in N and its spectrum equals the union of per-sector spectra to 10^{-10}; (iv) the analytic $V = 0$, 2×2, $N = 1$ spectrum $\{-2t, 0, 0, +2t\}$ (four-site ring); (v) the trace identity $\mathrm{Tr}\, H = V \sum$(occupied bonds); (vi) the $T \to 0$ and $T \to \infty$ limits of $\langle E \rangle$, S, and C_V. Ground-state energies ($V = 0$): $E_0/t = -2.828, -10.347, -18.000$ for $L = 2, 3, 4$. The full suite (57 tests) passes.

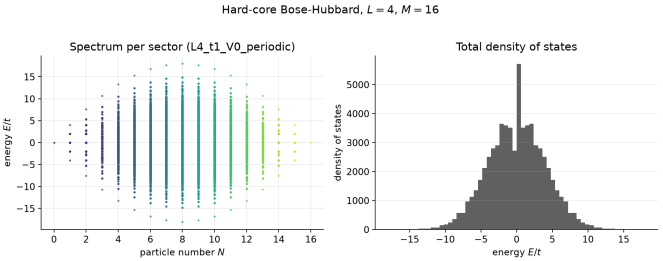


Figure 1: Eigenvalues per sector (left) and total density of states (right), $L = 4$, $V = 0$.

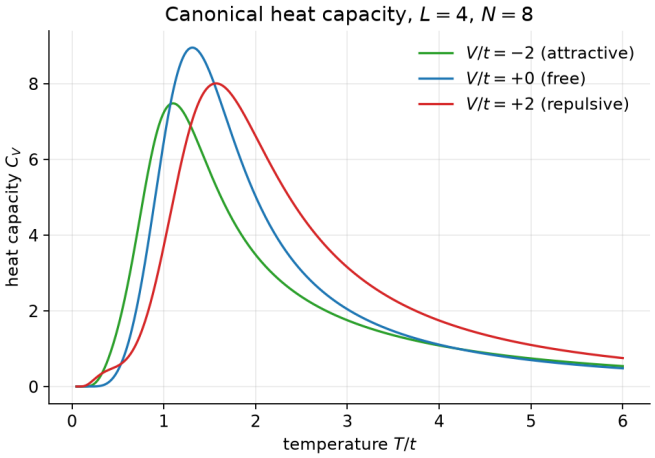


Figure 2: Canonical $C_V(T)$ at half filling ($L = 4$, $N = 8$) for free, repulsive, and attractive gases.

4 Results

4.1 Spectrum and density of states

Figure 1 shows the per-sector spectrum and total density of states for the $L = 4$ free gas. The bounded spectrum with a central DOS maximum is the structural origin of the negative-temperature region and of the finite-difference thermometry failure.

4.2 Heat capacity

The canonical heat capacity (Fig. 2) shows a single broad Schottky-like peak, vanishing as $T \to 0$ and $T \to \infty$. Repulsion shifts the peak to higher T, attraction to lower T, preserving the Schottky form.

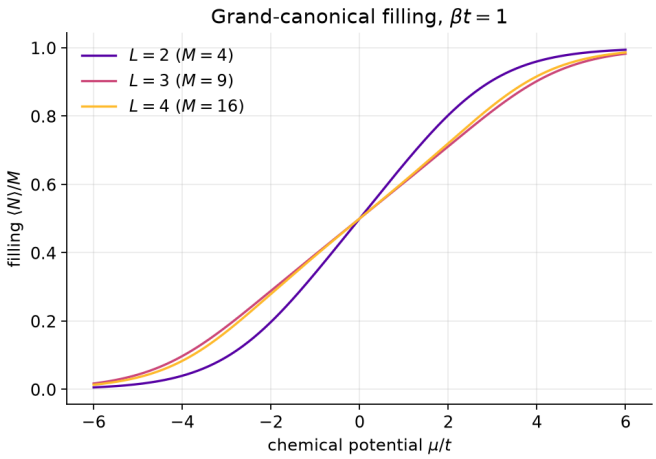


Figure 3: Grand-canonical filling versus μ/t at $\beta t = 1$, $L = 2, 3, 4$.

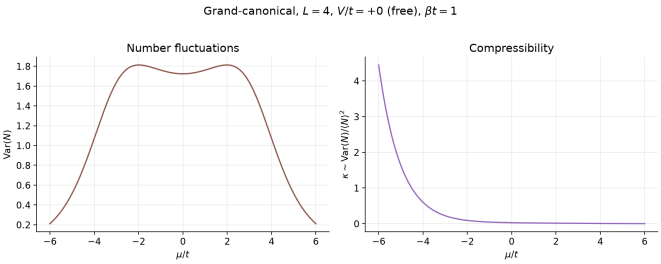


Figure 4: Number fluctuations and compressibility versus μ/t ($L = 4$, $V = 0$, $\beta t = 1$).

4.3 Grand-canonical filling and fluctuations

Figure 3 shows filling versus μ at $\beta t = 1$; all curves cross half filling at $\mu = 0$ (particle–hole symmetry), with $L = 3, 4$ converging toward a common thermodynamic profile while $L = 2$ deviates. Figure 4 shows $\mathrm{Var}(N)$ and the compressibility $\kappa \sim \mathrm{Var}(N)/\langle N \rangle^2$, peaking near half filling.

4.4 Ensemble energy comparison

Figure 5 overlays canonical and grand-canonical energy per site at matched μ^* (so $\langle N \rangle = N$). They coincide as $T \to 0$ and open a small gap at intermediate T—the finite-size inequivalence.

4.5 Entropy and the caloric curve

The canonical entropy per site $S/M = \beta \langle E \rangle / M + \ln Z_N / M$ (Fig. 6) rises from zero at $T \to 0$ (a non-degenerate ground state; the third law) and saturates at the infinite-temperature ceiling $\ln(\dim)/M$. Attraction raises the entropy fastest and repulsion slowest, mirroring the heat-capacity peak positions. Figure 7 makes the central methodological point graphically: the canonical caloric

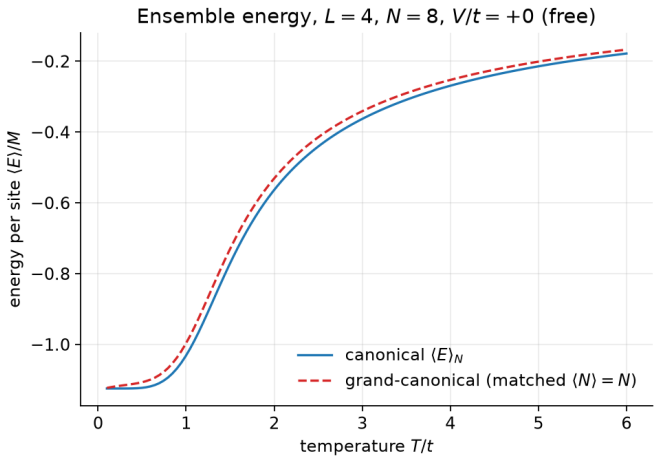


Figure 5: Canonical versus grand-canonical energy per site at matched particle number ($L = 4$, $N = 8$, $V = 0$).

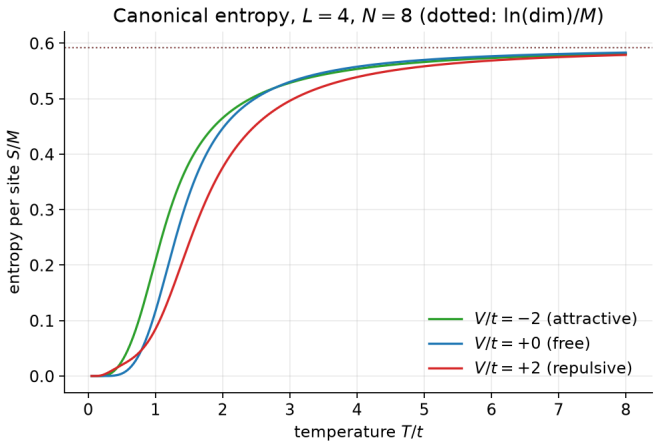


Figure 6: Canonical entropy per site S/M versus temperature ($L = 4$, $N = 8$); dotted lines mark the $T \to \infty$ ceiling $\ln(\dim)/M$.

curve $T(E)$ is smooth and monotone on the lower half of the spectrum, whereas the microcanonical finite-difference temperature $1/(\partial S/\partial E)$ scatters around it and, crucially, turns *negative* above the band center ($E \gtrsim 0$)—the bounded spectrum's density-of-states maximum, where $\partial S/\partial E$ changes sign. This is precisely the regime in which a single-energy microcanonical thermometer fails, and it justifies anchoring the equivalence analysis on fluctuation metrics rather than on $\partial S/\partial E$.

4.6 Finite-size scaling (central result)

Figure 8 and Table 1 present the three equivalence metrics versus $M = L^2$ with power-law fits. Two conclusions stand out. *(i) Equivalence emerges for free and repulsive gases*: all metrics decrease with size; the free-gas r_N and Δ_E/M follow clean power laws ($R^2 = 1.00$) with exponents between -0.6 and -0.9, bracketing the saddle-point expectation $M^{-1/2}$. *(ii) Equivalence fails*

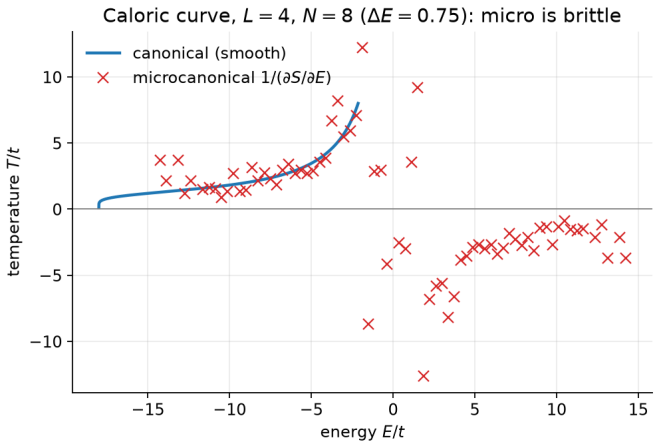


Figure 7: Caloric curve $T(E)$: smooth canonical (line) versus microcanonical finite differences (crosses) for $L = 4$, $N = 8$, $\Delta E = 0.75$. The microcanonical estimate is noisy and turns negative past the band center.

Table 1: Power-law exponents p (fit $\propto M^p$, $M = 4, 9, 16$) of each equivalence metric versus interaction, and the number fluctuation at the largest lattice $r_N(M{=}16)$. $\beta t = 1$, half filling, periodic. The r_N exponent rises monotonically and turns positive for strong attraction.

V/t	$p(r_E)$	$p(r_N)$	$p(\Delta_E/M)$	$r_N(M{=}16)$
-3	-0.92	$+0.13$	$+0.51$	0.962
-2	-0.88	-0.12	-0.59	0.553
-1	-0.91	-0.55	-0.37	0.240
0	-0.89	-0.64	-0.84	0.164
$+1$	-0.73	-0.65	-0.93	0.127
$+2$	-0.49	-0.69	-0.70	0.091
$+3$	-0.75	-1.10	-0.97	0.035

for the attractive gas in the number channel: the r_N decay exponent varies monotonically with interaction (Table 1), from $p = -1.10$ at $V/t = +3$ (strong repulsion restores equivalence fastest) through -0.64 at $V = 0$, up to $p = +0.13$ at $V/t = -3$. The sign change is the key result: for strong attraction the relative number fluctuation at $M = 16$ reaches 0.96 and its exponent turns *positive*, so the canonical–grand mismatch *grows* with system size rather than shrinking. This is the fingerprint of a developing clustering/phase-separation instability: the grand ensemble supports large number fluctuations between competing fillings that the fixed-N canonical ensemble forbids, and the effect strengthens with size. The energy metrics (r_E, Δ_E/M) still shrink for all V—the inequivalence is specific to the particle-number channel, where the instability lives.

4.7 Interaction dependence

Figure 9 collects the metrics at $L = 4$ versus V/t over seven couplings. A still denser sweep is cheap at fixed $L = 3$: Fig. 10 resolves the full trend over 33 values of V/t. The relative number fluctuation $\sigma_N/\langle N\rangle$ rises smoothly and monotonically toward unity as the interaction becomes more attractive—the continuous form of the canonical–grand breakdown driven by clustering—while the relative energy fluctuation grows instead with repulsion. The per-site energy gap stays small and develops shallow minima. The anomaly is thus specific to the number channel and to attraction, confirming it as a genuine physical effect rather than a fitting artifact.

5 Discussion

Why fluctuations, not thermometry. The canonical distribution is the Laplace transform of the microcanonical density of states, and equivalence is the statement that this transform is saddle-point dominated. The width $\sigma_E/|\langle E\rangle|$ measures saddle sharpness directly; it is a smooth spectral sum, free of the differentiation noise that cripples $\partial S/\partial E$ on a discrete spectrum. The same holds for $\sigma_N/\langle N\rangle$. The fluctuation metrics are thus both robust and correct.

The attractive anomaly. An attractive nearest-neighbor interaction lowers the energy of clustered configurations, driving a low-temperature tendency toward phase separation. In the grand ensemble this yields a broad $P(N)$ with large $\mathrm{Var}(N)$ that does not self-average at these sizes; the canonical ensemble, with N pinned, cannot access the competing fillings, so the ensembles disagree. This is the finite-size precursor of the thermodynamic failure of equivalence at first-order transitions.

Limitations. With only $M = 4, 9, 16$, the exponents are indicative, not precise, and the $M = 4$ point leverages the fits—doubly so because the $L = 2$ periodic lattice de-duplicates to a single-bonded four-site ring rather than the true double-bonded two-torus (its $N = 1$ band is $\{-2t, 0, 0, 2t\}$ instead of the dispersion minimum $-4t$), so it is a slightly different lattice from the $L = 3, 4$ members of the sequence. The $L = 3, 4$ single-particle spectra reproduce the exact tight-binding band $-2t[\cos k_x + \cos k_y]$ to machine precision. Results are at a single temperature $\beta t = 1$; the microcanonical T, μ are reported only as a window-sensitive diagnostic; the bounded spectrum admits negative-temperature states that the fluctuation metrics sidestep. The dense interaction sweep of Fig. 10 is computed at fixed $L = 3$ (open boundaries), where many V/t values are inexpensive.

6 Conclusion

Using numerically exact diagonalization of the 2D hard-core Bose–Hubbard model, we gave a transparent, reproducible account of how ensemble equivalence emerges with system size in an interacting quantum lattice gas. By recognizing the ill-conditioning of finite-difference microcanonical thermometry and instead measuring equivalence

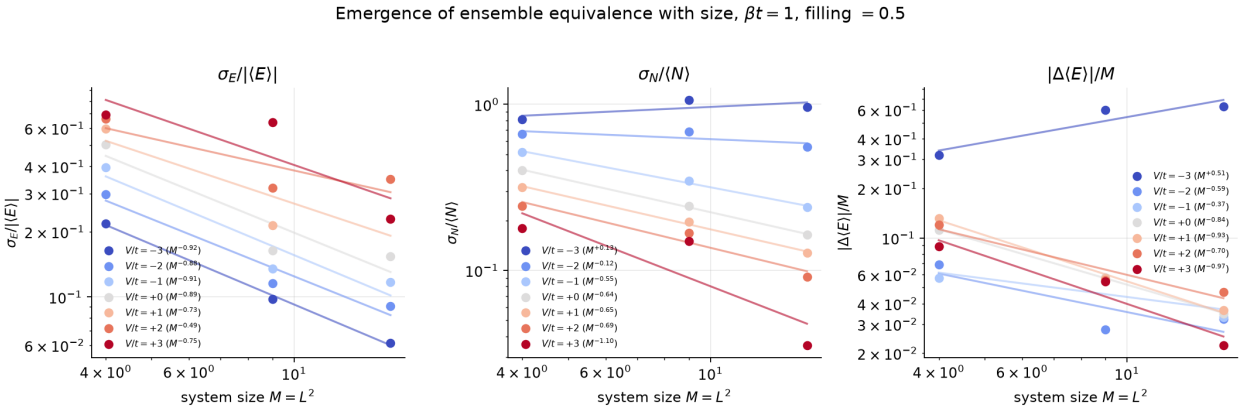


Figure 8: Finite-size scaling of r_E, r_N, and Δ_E/M for seven couplings $V/t \in \{-3, \ldots, +3\}$ (attractive blue, repulsive red) at $\beta t = 1$, half filling. Lines are power-law fits.

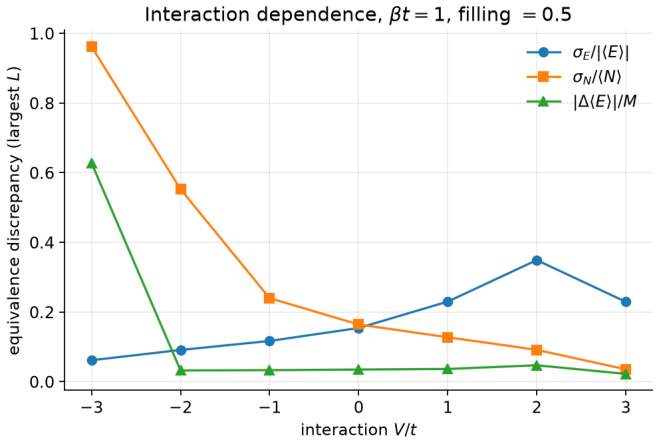


Figure 9: Equivalence metrics at $L = 4$ versus V/t (seven couplings). The number fluctuation $\sigma_N/\langle N\rangle$ falls monotonically from $\sim$1 (strong attraction) to $\sim$0 (strong repulsion).

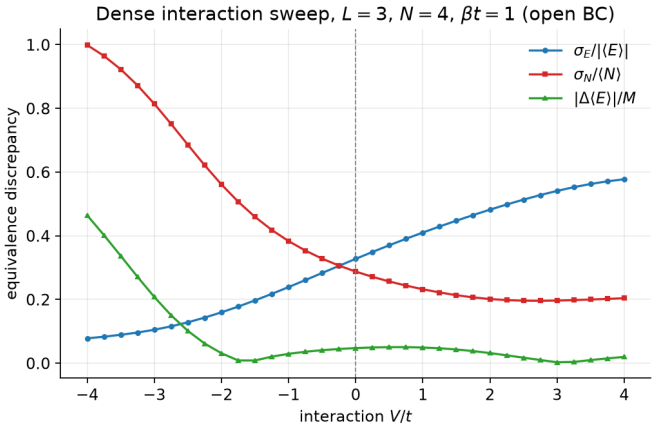


Figure 10: Dense interaction sweep at fixed $L = 3$ (33 couplings, open BC, $\beta t = 1$): $\sigma_N/\langle N\rangle$ grows toward unity with attraction, the number-channel signature of ensemble inequivalence.

through robust fluctuation observables, we demonstrated monotone power-law convergence for the free and repulsive gases and a clear breakdown of canonical–grand equivalence in the number channel for the attractive gas—the finite-size signature of its phase-separation instability.

Author contributions. P. Das, A. Shakya, and D. Thakur conceived the study, developed the exact-diagonalization and ensemble framework, performed the calculations and analysis, and wrote the manuscript.

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